

Preliminary Design Review

AERO62520 Robotic Systems Design Project

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1 Introduction

The project aims to design and build an autonomous robot capable of detecting and retrieving a specified object from its environment without human intervention. The formal statement of the problem for group 2 is to design, implement, and validate a robot that can independently retrieve a target object from an internal environment.

To meet this challenge, group 2 proposes a mobile manipulation system that combines a Leo Rover mobile base with a myCobot 280 Pi robotic arm mounted on board. This platform is augmented with key perception sensors, including a rplidar a2m12 and an Intel realsense depth camera, as well as hardware for computing and communication. Together, these subsystems are designed to allow autonomous navigation, object detection and manipulation in an initially unmapped environment.

This preliminary design review illustrates the overall architecture of the system, the hardware and software choices and the integration strategies that will be used to transform the Leo rover standard platform into a complete robot capable of meeting the agreed performance and safety requirements.

2 System

The components required for constructing the system are outlined below. Each element plays a specific role in ensuring the overall functionality and performance of the design. The interconnections and data flow between these components are clearly illustrated in Fig. 1, providing a comprehensive overview of how the system operates as an integrated whole.”

2.1 Battery

Provides DC power to the whole robot. It supplies the Leo core board, the Raspberry Pi, the Intel NUC, the sensors, and the manipulator.

2.2 Leo Core Board

Embedded control board of the Leo Rover. It drives the four DC motors, reads the wheel encoders, monitors battery status, and exposes low-level interfaces to the Raspberry Pi (e.g. velocity commands and odometry).

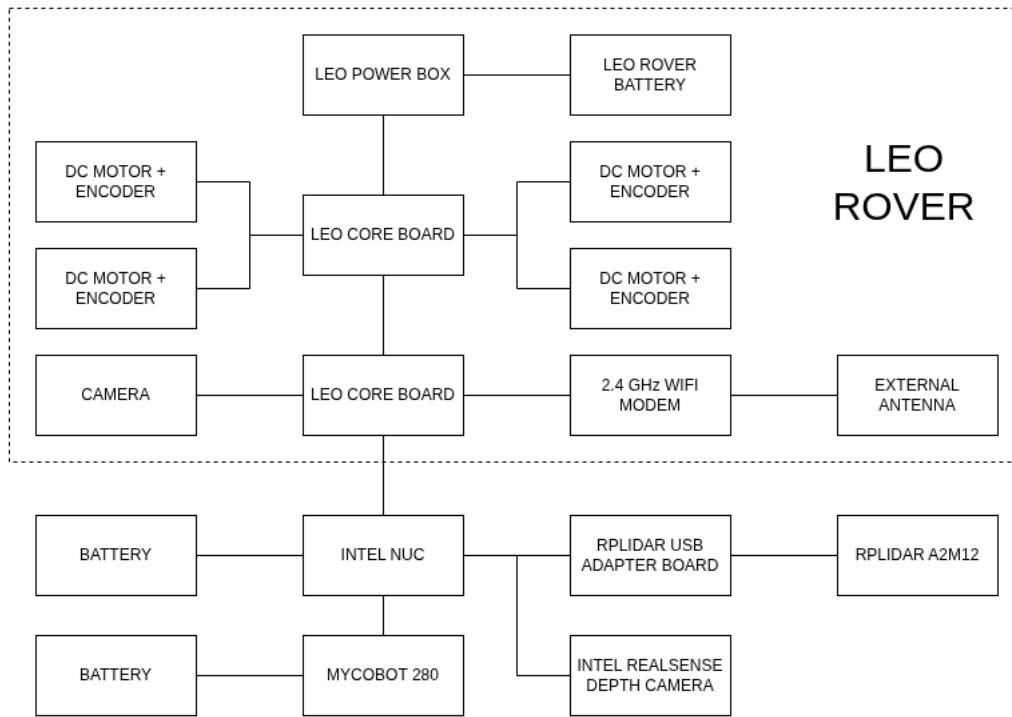


Fig. 1. System block diagram

2.3 DC Motors + Encoders

Each motor drives one wheel of the rover. The encoders measure wheel rotation, allowing the system to estimate the linear and angular velocity of the base and to provide odometry for navigation.

2.4 Raspberry Pi 4 Model B

On-board computer dedicated to low-level control of the Leo Rover. It communicates with the Leo core board, runs the basic ROS nodes for the base, and forwards sensor and odometry data to the Intel NUC. It also receives high-level velocity commands from the NUC and sends them to the motor controller.

2.5 2.4 GHz Wi-Fi Modem + External Antenna

Provides wireless network connectivity between the robot and the lab network / operator laptop. It allows remote login, code deployment and monitoring, and enables ROS communication between the robot and external devices if needed.

2.6 Camera

Forward-facing RGB camera mounted on the rover. It is mainly used for teleoperation, visual monitoring and optional visual feedback during navigation and experiments.

2.7 Intel RealSense Depth Camera

RGB-D camera is used to detect and localise the target wooden block in 3D. The depth information allows the system to estimate the distance and orientation of the object and to provide precise pose information to the manipulator for grasping.

2.8 Intel NUC

Main on-board computer and “brain” of the system. It runs the high-level ROS2 stack, including SLAM, global and local navigation, behaviour tree or state-machine control, object detection, grasp planning and coordination between the mobile base and the manipulator.

2.9 RPLIDAR USB adapter board + RPLIDAR A2M12

The RPLIDAR A2M12 is a 2D scanning LiDAR that produces 360° range measurements around the rover. Through the USB adapter board it connects to the Raspberry Pi 4. Its data are used for mapping (SLAM), obstacle detection and collision-free path planning.

2.10 ER myCobot 280 Pi

Six-degree-of-freedom robotic arm mounted on the rover. It receives target poses from the perception system and executes pick-and-place motions to grasp the wooden block and place it at the retrieval location, completing the object-retrieval task

3 Mechanical Design

3.1 3D CAD Model and System Overview

The mechanical design integrates a 6 DOF robotic manipulator onto the mobile chassis using a custom designed payload sled. This configuration maximizes the operational workspace while maintaining

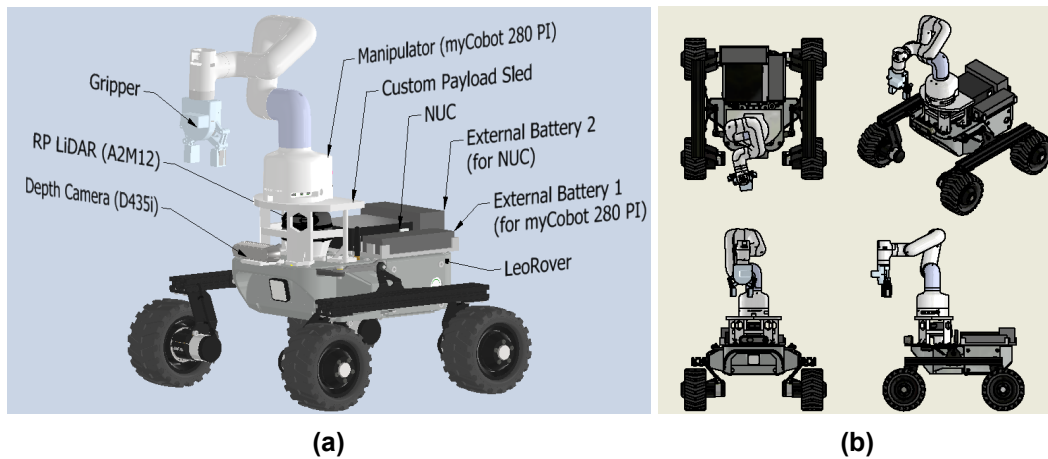


Fig. 2. Mobile manipulator platform based on LeoRover. (a) Annotated isometric view identifying key components. (b) Orthographic and isometric views showing the overall geometry.

navigational capabilities. This section addresses the Mechanical Design requirements of the PDR, including the 3D CAD model of the robot, the payload sled design files, and an analysis of component placement trade-offs, manufacturability, and structural suitability.

As illustrated in Fig. 2, the system features a myCobot 280 PI manipulator mounted on an elevated deck. The RPLidar A2M12 and RealSense D435i are strategically positioned to ensure unobstructed fields of view for autonomous navigation and obstacle avoidance.

3.2 Payload Sled Design Files

The payload sled serves as the central interface, connecting the manipulator and sensors to the rover chassis. A modular design approach was adopted to facilitate assembly and maintenance. Fig. 3 demonstrates the vertical stacking logic. The sled is composed of discrete modules, including the Manipulator Base (Item 10) and sensor / external battery / NUC mounts (Items 2, 4, and 6). The exploded view highlights the assembly axis (indicated by blue trail lines), showing how the 3D printed struts and plates interlock with the chassis via standard M5 fastening points. This modularity ensures that individual components can be rapidly replaced without discarding the entire assembly. (<https://a360.co/4pViC3t>)

3.3 Component Placement Trade-offs

The arrangement of hardware components involves a critical balance between sensor visibility, mechanical interference, and system stability.

- **Sensor Blocking vs. Mechanical Clearance**

As illustrated in Fig. 4 (a) and (b), the primary design challenge was to prevent the sensors

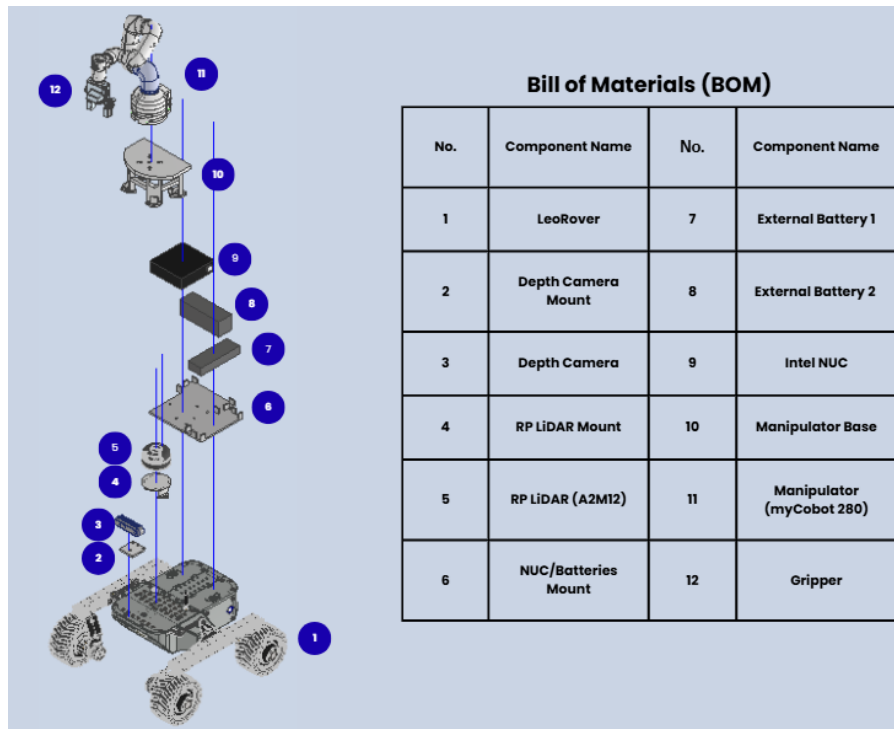


Fig. 3. Exploded view of the payload sled assembly with Bill of Materials (BOM). (Standard M5 fasteners are omitted from the exploded view for visual clarity. The fastening strategy is described in Section 3.4.)

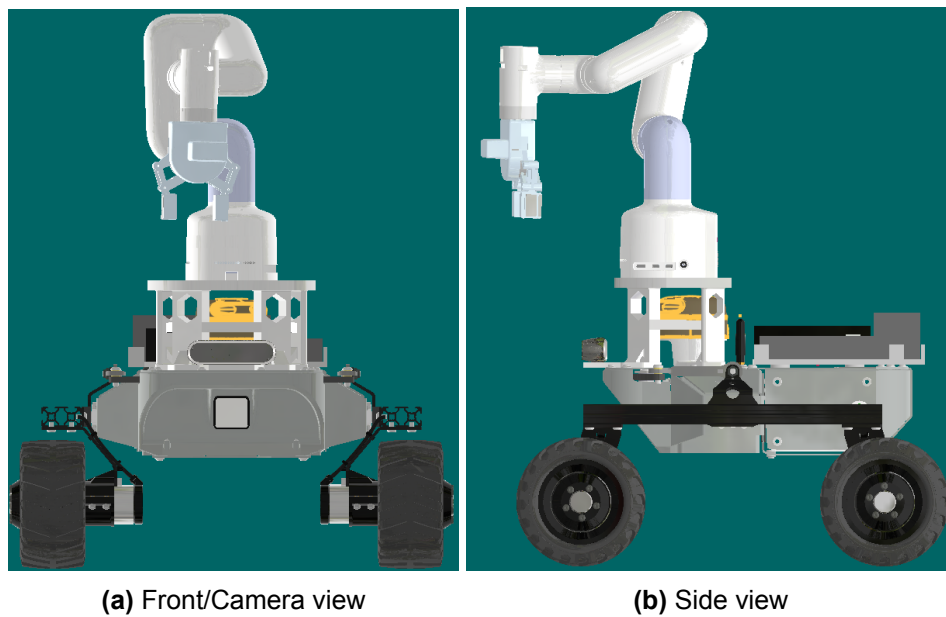


Fig. 4. Component placement analysis. (a) Front view showing sensor alignment; (b) Side view illustrating the elevated sensor stack and manipulator.

(LiDAR and Depth Camera) from obstructing each other or being blocked by the chassis components.

- **LiDAR Elevation**

The RPLidar is mounted on a raised base. This elevation is strictly necessary to clear the height of the NUC unit positioned at the rear of the rover, ensuring an unobstructed 360-degree Field of View for navigation.

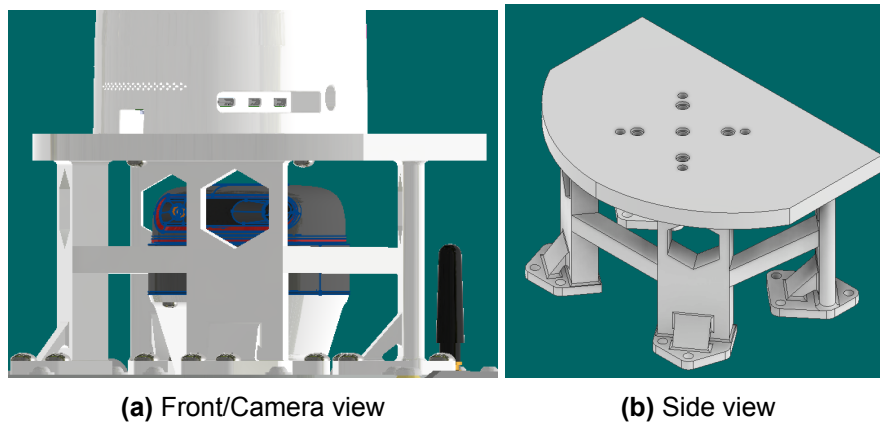


Fig. 5. (a) & (b): Detailed view of the custom manipulator base featuring hexagonal cutouts for LiDAR visibility.

- **Manipulator Height Trade-off**

A direct consequence of raising the LiDAR is the requirement to elevate the robotic manipulator. If the arm were mounted at the chassis level, its movement would be physically obstructed by the raised LiDAR mount. Therefore, the manipulator was elevated via the custom sled. While this increases the system's total height, it is a necessary sacrifice to resolve the interference between the navigation sensors and the manipulator's workspace.

- **Center of Gravity Mitigation**

To balance this top-heavy configuration, the heaviest components (the Battery and NUC) are positioned at the lowest section of the rear chassis to act as counterweights, ensuring dynamic stability.

- **Sled Geometry Optimization**

The payload sled was meticulously designed as seen in Fig. 5 to resolve the conflict between the LiDAR's performance and the arm's reach, while also providing a stable structural foundation to support the robotic manipulator mounted above.

- **Vertical Clearance**

The platform height provides a minimum clearance of **30 mm** above the LiDAR's scanning plane. This margin accounts for the RPLidar's vertical beam divergence, ensuring the laser is not clipped by the plate. However, the height is strictly capped; raising it further would compromise the center of gravity and prevent the arm from reaching lower targets (e.g., objects on boxes).

- **Hexagonal Cutouts**

To minimize blind spots in the LiDAR's data, hexagonal cutouts were engineered into the support struts. This reduces the structural cross-section obstructing the 360° scan while retaining sufficient material strength to support the manipulator's dynamic loads.

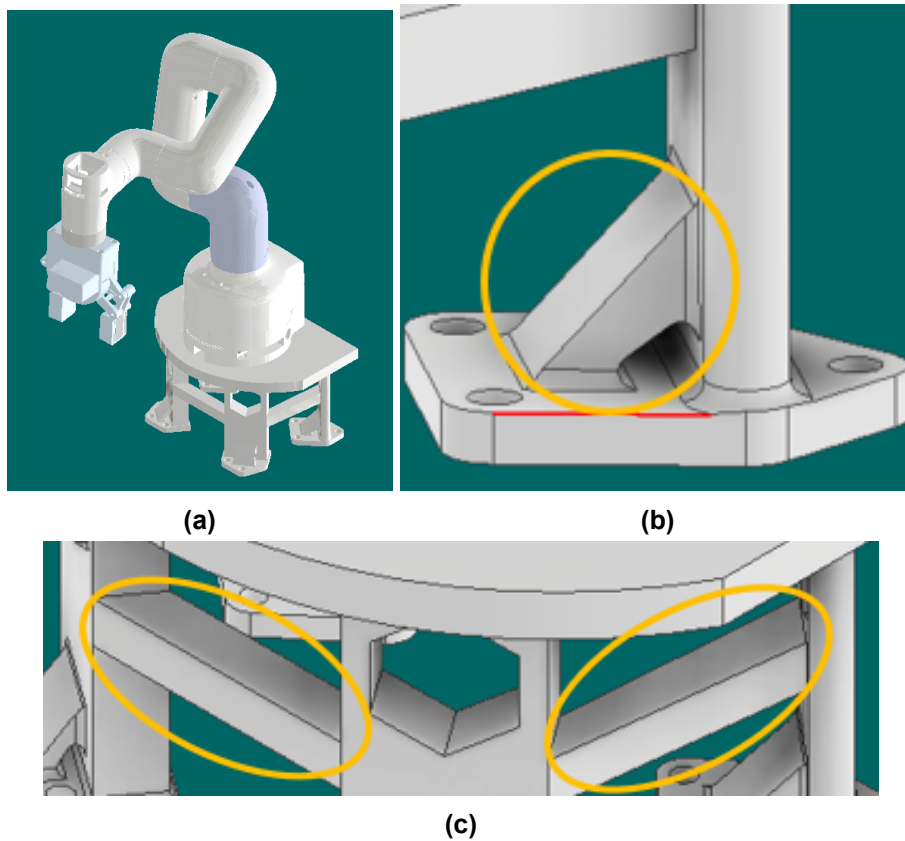


Fig. 6. Structural optimization details. **(a)** Isometric view showing the manipulator integrated onto the sled. **(b)** Reinforced base struts featuring triangular gussets. **(c)** Transverse cross-beams connecting the struts to enhance torsional rigidity.

Overall, the final component layout resolves the conflicts between sensor visibility, manipulator reach, and system stability while staying within the physical constraints of the LeoRover chassis.

3.4 Manufacturability and Suitability Analysis

The payload sled design was optimized, as shown in Fig. 6, for FDM 3D printing (PLA/PETG), prioritizing structural rigidity while adhering to time-and-cost-efficient manufacturing principles.

3.4.1 Manufacturability (Design for Additive Manufacturing)

To minimize production costs and lead time, the design strictly follows DfAM (Design for Additive Manufacturing) guidelines:

- **Print Orientation & Self-Supporting Geometry:** As verified in the slicer software (Bambu Studio), the sled is designed to be printed in an inverted orientation. The hexagonal cutouts and angular struts feature overhangs between 45° and 60° , allowing them to be self-supporting

- **Minimal Support Requirement:** This orientation eliminates the need for large internal support structures for the top plate. Support material is restricted solely to the small surface area of the mounting feet overhangs, significantly reducing material waste.
- **Tolerance and Print Settings:** A dimensional tolerance of +0.2mm was applied to all mounting holes to account for thermal shrinkage, ensuring a seamless fit for fasteners. The parts are specified to be printed with 4 perimeter walls and 40% gyroid infill. This specific infill pattern provides isotropic strength, maximizing the load-bearing capacity while minimizing weight.

3.4.2 Suitability and Structural Integrity

The structural design ensures stability under the dynamic loads of the manipulator, utilizing a reinforced frame architecture:

- **Z-Axis Reinforcement (Figure 3.5b):** Triangular reinforcement ribs were integrated into the base of the struts. These ribs distribute bending moments over a larger area, mitigating the risk of layer delamination, the most common failure mode in FDM parts loaded laterally.
- **Lateral Stability (Figure 3.5c):** To prevent structural wobbling during arm acceleration, transverse cross-beams were added to connect the vertical struts. These beams create a rigid frame that resists lateral deflection and torsional forces.

3.5 Justification of Structural Analysis (Fit for Purpose)

Several aspects are used as parameters to analyze and justify the structural design of the system. Those aspects are listed below,

- **Load Analysis:** Given the low total static load applied to the sled, which comprises the myCobot 280's net weight (850 g) and its maximum payload capacity (250 g), the resulting downward force is approximately 1.1 kg (11 N). Relative to the sled's structural volume, cross-beam reinforcement, and high infill density, this load is minimal.
- **Chassis Connection Interface Analysis:** The payload sled is rigidly secured to the Leo Rover's aluminum mounting deck using ten M5 socket head cap screws (see Fig. 7.). The use of M5 fasteners is dictated by the rover's standard mounting interface rather than load requirements. A single Grade 8.8 M5 bolt offers a shear strength exceeding **3,000 N**. With 10 connection points, the total shear resistance exceeds **30,000 N**. Compared to the manipulator's dynamic load (<

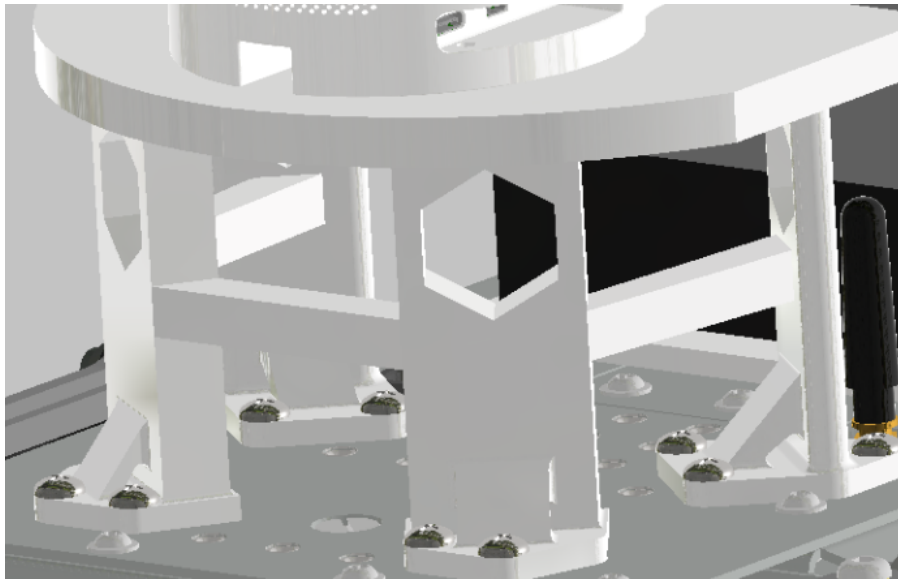


Fig. 7. Close-up view of the sled base secured to the Leo Rover chassis using ten M5 fasteners.

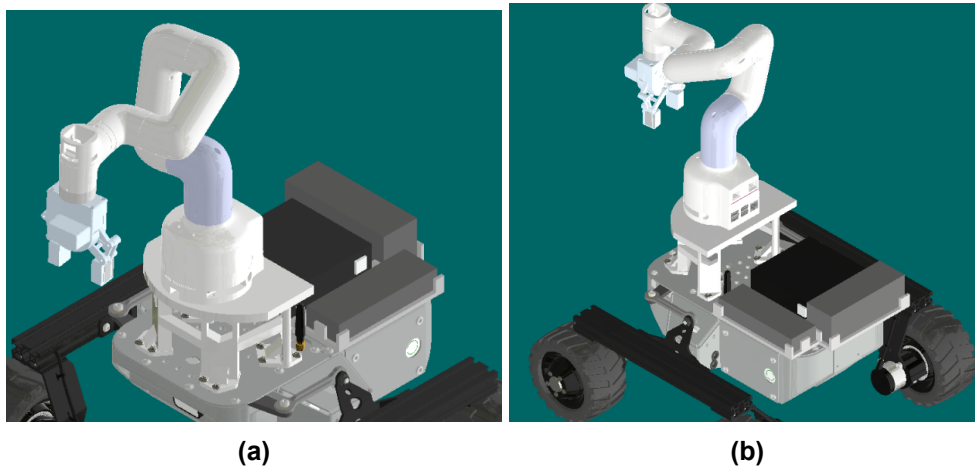


Fig. 8. (a) and (b): Rear mounted battery units acting as counterweights to offset the manipulator's reach.

50 N), this geometric constraint results in a massive safety factor (>600), rendering the interface virtually fail-proof.

- **System Stability & Mass Distribution Strategy:** To ensure operational stability, the power configuration serves a dual mechanical purpose. As shown in Fig. 8, two auxiliary battery units are mounted on the rear chassis deck. This strategic placement shifts the global Center of Gravity rearward, creating a restoring moment. This effectively negates the forward tipping torque generated when the manipulator is fully extended, ensuring passive stability during pick-and-place operations without the need for active stabilization control.
- **Conclusion: Engineering judgment** indicates that the current design provides a safety factor well beyond operational requirements for structure, connection, and stability. Therefore, full Finite Element Analysis (FEA) or dynamic tipping simulations would offer diminishing returns and were considered unnecessary for this PDR, in line with efficient engineering practice.

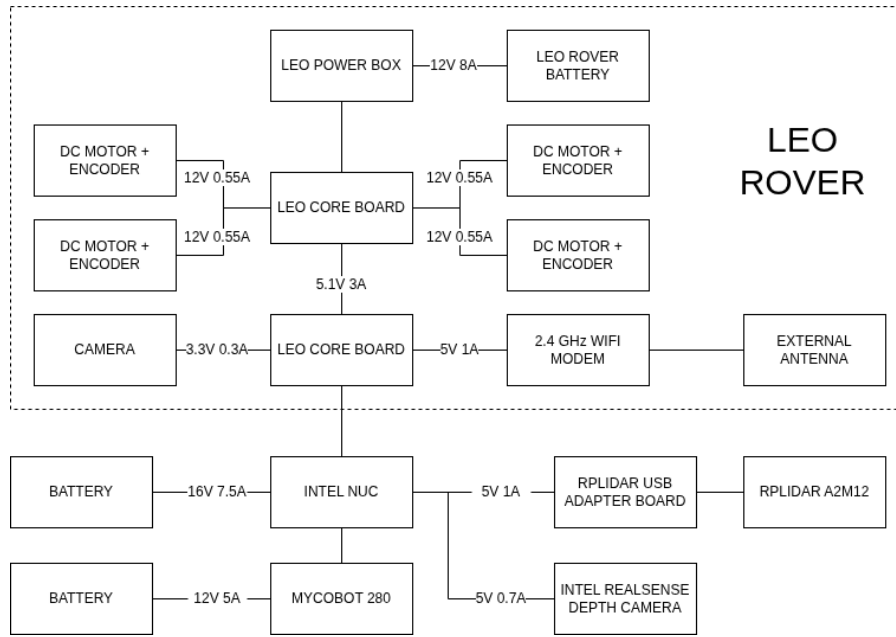


Fig. 9. Power distribution diagram

4 Electrical Design

According to the Leo Rover specification this is a battery with a nominal voltage of 12 V and a maximum current of 8 A for the whole rover, which corresponds to roughly 100 W of continuous power. The proposed system will use two additional batteries to power other peripherals. The first battery, which powers the NUC, has a nominal voltage of 14.8 V, a capacity of 6700 mAh, and a maximum discharge rate of 65 C. This is sufficient to supply power to the NUC, which requires up to 120 W at peak load with an operating voltage range of 12–20 VDC. The second battery is used to power the myCobot 280. According to its specifications, the robot requires 60 W of power at 12 V. To meet this requirement, a 3S LiPo battery is used, featuring a capacity of 4500 mAh, a nominal voltage of 11.1 V, and a maximum discharge rate of 55 C. The voltage of this battery ranges from 2.7 to 4.2 V per cell, depending on its remaining capacity. Based on calculations shown in equations 1-6, the discharge rate for these batteries will be approximately 1.2 C, which is significantly lower than their maximum limit. Furthermore, these batteries are sufficient to power the system for more than 30 minutes, as required. The power distribution of this system can be seen in Fig. 9.

Power Calculation for NUC:

$$Energy = Capacity \times Voltage = 6700mAh \times 14.8V = 99.16Wh \quad (1)$$

$$Duration = \frac{BatteryEnergy}{DevicePower} = \frac{99.16Wh}{120W} \approx 0.826hours \approx 49.6minutes \quad (2)$$

$$I = \frac{P}{V} = \frac{120W}{14.8V} = 8.1A \quad (3)$$

Power Calculation for myCobot280:

$$Energy = Capacity \times Voltage = 4500mAh \times 11.1V = 49.9Wh \quad (4)$$

$$Duration = \frac{BatteryEnergy}{DevicePower} = \frac{49.9Wh}{60W} \approx 0.832hours \approx 49.9minutes \quad (5)$$

$$I = \frac{P}{V} = \frac{60W}{11.1V} = 5.4A \quad (6)$$

5 Software Design

This section presents the software architecture of our mobile manipulator for autonomous corridor navigation and block pick-and-place. The software is implemented in ROS 2 and distributed across three compute units: an Intel NUC mounted on the mobile base, a Raspberry Pi 4 on the Leo Rover platform, and the embedded Raspberry Pi inside the myCobot 280 Pi arm. High-level perception, planning and task coordination run on the NUC, while low-level control and hardware interfaces run on the Raspberry Pi boards. All nodes communicate over a shared Wi-Fi network using ROS 2 topics, services and actions. Here is the link to the github repository https://github.com/Peanutman917/Robotics-Systems-Design-Project-Group2_PGD_UOM

5.1 Software architecture overview

The overall software architecture is organised around four main functional modules: perception, navigation and mapping, task coordination, and low-level control of the base and arm.

On the Intel NUC, we run the perception node, the task coordinator and the navigation stack together with the map / costmap server. The RealSense D435i camera publishes colour (and aligned depth) images on topics such as /camera/color/image_raw, which are consumed by the perception node to detect coloured blocks and estimate their 3D positions. The navigation stack and map / costmap server use LiDAR scans and wheel odometry to perform localisation and path planning, and publish velocity commands on /cmd_vel.

On the Raspberry Pi 4 at the Leo Rover base, we run the LiDAR driver node, the motor / encoder drivers and the base controller node. The LiDAR driver node reads data from the RPLIDAR A2M12 sensor and publishes 2D laser scans on /scan. The motor / encoder drivers interface with the four wheel motors and encoders, publishing wheel odometry. The base controller node subscribes to /cmd_vel from the NUC, converts these commands into individual wheel speed commands and drives the wheel motors accordingly.

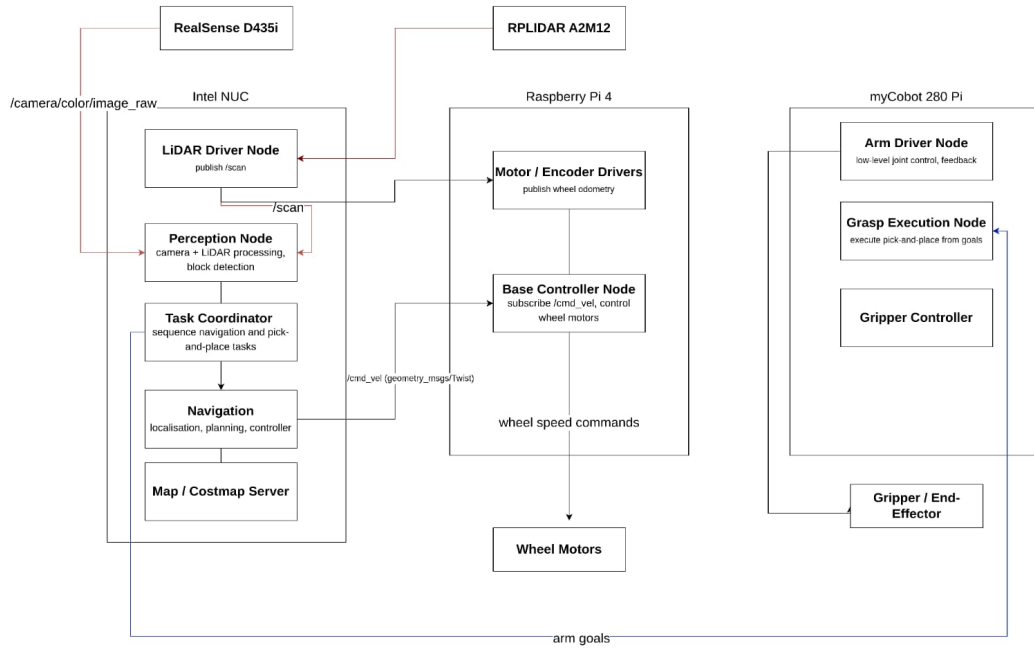


Fig. 10. Software architecture and node deployment across three computers

On the myCobot 280 Pi controller, we run the arm driver node, the grasp execution node and the gripper controller. The arm driver node provides low-level joint control and feedback for the myCobot arm. The grasp execution node receives pick-and-place goals from the task coordinator and generates the corresponding arm motions. The gripper controller opens and closes the gripper around the detected object and releases it at the drop location.

Fig. 10 shows the software architecture and node deployment across the three compute units, together with the main sensors and actuators they interface with.

5.2 Software block diagram

The same architecture can be represented as a software block diagram that emphasises the main ROS 2 nodes and the data flow between them. On the Intel NUC, the perception node subscribes to the RealSense D435i colour and depth image topics, and may also subscribe to `/scan` when LiDAR information is needed for obstacle or corridor awareness; however, LiDAR data is not the primary input for block detection. The perception node outputs detected block poses, which are consumed by the task coordinator.

The task coordinator selects the next target block and dispatches two classes of commands: navigation goals to the navigation stack and grasp goals to the grasp execution node on the myCobot controller. The navigation stack and map / costmap server subscribe to `/scan`, wheel odometry and

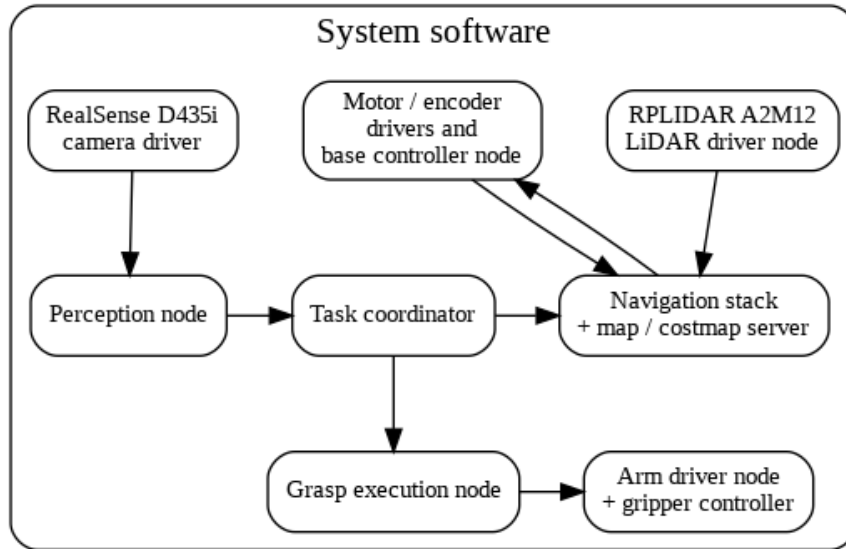


Fig. 11. Software block diagram

the TF tree to maintain global and local costmaps, compute collision-free paths and generate velocity commands on `/cmd_vel`.

On the Raspberry Pi 4, the LiDAR driver node publishes `/scan` and the motor / encoder drivers publish wheel odometry. The base controller node subscribes to `/cmd_vel` and produces motor-level wheel commands. This separation ensures that time-critical control remains local to the base while computationally intensive planning and coordination remain on the NUC.

On the myCobot 280 Pi controller, the grasp execution node receives grasp and place goals, plans the motion sequence (approach, grasp, lift, move, place and retreat), and sends joint commands to the arm driver node. The gripper controller is triggered in synchrony with these phases to close during the grasp phase and open at the drop location.

Fig. 11 illustrates this software block diagram and the main topics connecting the perception, navigation, base control and manipulation subsystems.

5.3 Perception and object detection pipeline

The perception node on the Intel NUC implements an RGB-D pipeline that converts RealSense data into a list of detected blocks with estimated 3D positions in the robot base frame as shown in Fig. 12. The pipeline is designed to be lightweight and reliable for the limited set of target colours and shapes required by the corridor pick-and-place task.

First, the input image is pre-processed by cropping a region of interest and applying basic noise filtering. Colour segmentation is then applied to isolate candidate regions for each target colour under

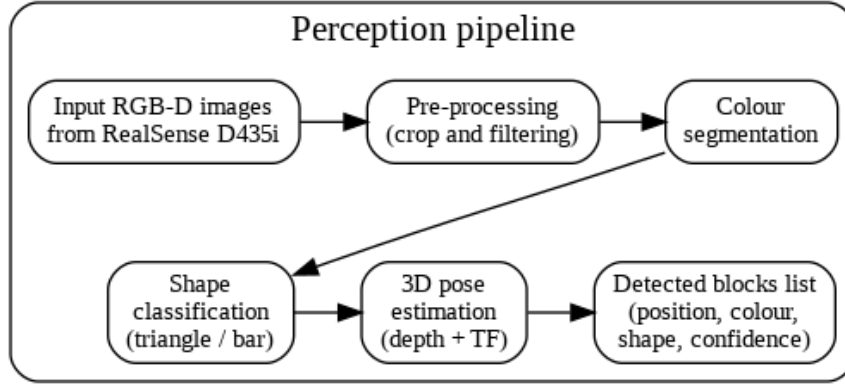


Fig. 12. Perception pipeline diagram

indoor lighting. For each colour candidate, the primary shape classification method uses contour-based geometric features to distinguish triangular and bar-shaped blocks. This choice keeps inference fast and explainable for the PDR stage. As a secondary option, if contour-based classification becomes unreliable under extreme viewpoint changes, partial occlusions or lighting variation, a lightweight YOLOv8 detector can be introduced for end-to-end detection and class assignment. The YOLOv8 output can be used to refine the 2D region of interest and improve robustness before depth-based 3D estimation.

For each accepted candidate region, the aligned depth image is used to back-project the pixel region into a 3D point in the camera frame. The pose is then transformed to the robot base frame using TF. Here, TF (tf2) provides the coordinate transformations between the camera frame, base frame and odometry/map frames, ensuring that perception, navigation and manipulation share a consistent spatial reference. The perception node then publishes a structured detection list that includes position, colour, shape and confidence, which the task coordinator uses for target selection and grasp planning.

5.4 Navigation and base control

Navigation and base control are implemented jointly by the navigation stack and map / costmap server on the Intel NUC and the LiDAR driver node, motor / encoder drivers and base controller node on the Raspberry Pi 4. The LiDAR driver node publishes /scan from the RPLIDAR A2M12, and the motor / encoder drivers provide wheel odometry required for localisation.

On the NUC, the navigation stack subscribes to /scan, odometry and TF to maintain global and local costmaps of the corridor environment and to compute collision-free paths to target poses. It outputs a stream of velocity commands on /cmd_vel. When the task coordinator requests movement to a search area, a grasping pose or the drop zone, it issues a navigation goal to the navigation stack; the resulting motion is automatically executed by the base controller.

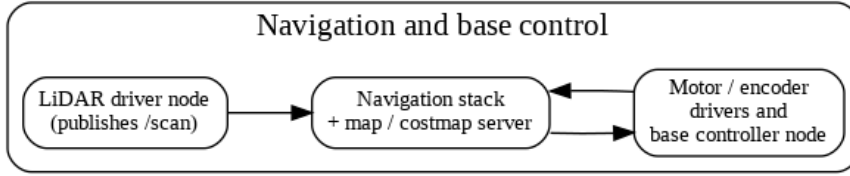


Fig. 13. Navigation and base control

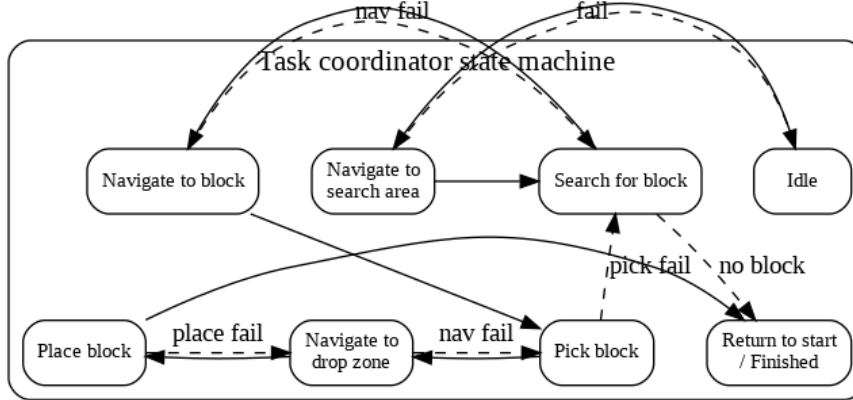


Fig. 14. FSM of the task coordinator with success and recovery transitions

On the Raspberry Pi 4, the base controller node receives `/cmd_vel` and converts these commands into wheel speed signals for the four motors. This architecture ensures that low-latency motor control remains on the embedded platform while higher-level planning and localisation remain on the NUC. The consistent TF tree shared across navigation, perception and manipulation also guarantees that block poses estimated from the camera can be used directly for navigation and grasping without frame ambiguity.

5.5 Manipulation and task coordination

Manipulation and task coordination are implemented by the task coordinator on the Intel NUC and the arm driver node, grasp execution node and gripper controller on the myCobot 280 Pi controller. The task coordinator is structured as a finite state machine (FSM) to ensure predictable behaviour and robust recovery across the full pick-and-place cycle.

The nominal mission sequence begins in `Idle`, then transitions through `Navigate to search area`, `Search for block`, `Navigate to block`, `Pick block`, `Navigate to drop zone`, `Place block`, and finally `Return to start / Finished`. The task coordinator waits for stable perception outputs before committing to a target and then sends the selected block pose in the base frame to the grasp execution node. The grasp execution node generates an arm motion sequence (approach, grasp, lift, move, place and

retreat), commands the arm driver node accordingly, and synchronises the gripper controller to close during the grasp phase and open at the drop location.

Failure transitions are explicitly defined for navigation failures, missing detections and unsuccessful grasps. In these cases, the FSM returns to earlier safe states such as Search for block or Navigate to block, rather than leaving the system in an undefined condition. This improves robustness and simplifies system-level debugging during integration testing.

6 Analysis

Table 1. Requirements - Functional Table

Require- ment no.	Source	Shall statement	Verification success criteria	Analysis/Current status
1.1	Requirements- Functional	The robot shall include a forward-facing RGB camera and publish '/camera/im- age_raw' at ≥ 10 Hz with resolution ≥ 640Ö480 calibration loads on startup	1) Startup log shows the calibration file loaded without errors. 2) Spot-check 10 frames — no tearing or major drops.	1.The initial system startup confirms that the D435i depth camera is detected and its configuration loads successfully with no calibration errors. 2.The review shows stable RGB and depth output with no noticeable tearing, artefacts, or frame drops.

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Requirement no.	Source	Shall statement	Verification success criteria	Analysis/Current status
1.2	Requirements-Functional	The robot shall include a 2D LiDAR that publishes 'scan' at $\geq 5\text{Hz}$ with an effective indoor range of $\approx 0.15\text{--}3.5\text{ m}$ for mapping and obstacle detection.	1) Display 'scan' for 60 s; average rate $\geq 5\text{ Hz}$ with no timeouts. 2) Validate ranges at 1 m and 2 m against a ruler/board (error $\leq \pm 10\text{ cm}$). 3) Generate an occupancy grid in a static layout and visually match obstacles	1.The /scan topic was monitored for 60 seconds, showing a stable average rate above 5 Hz with no timeouts, confirming consistent LiDAR data publishing. 2.Range checks at 1 m and 2 m against a ruler/board showed errors within $\pm 10\text{ cm}$, meeting the required accuracy tolerance. 3.The generated occupancy grid correctly represented the static environment, with mapped obstacles visually matching their real-world positions, demonstrating reliable environment perception.

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Require- ment no.	Source	Shall statement	Verification success criteria	Analysis/Current status
1.3	Requirements- Functional	The robot shall include an on-board computation device capable of launching perception, control, and teleoperation nodes via a single command and running for ≥ 5 minutes without crash.	1) Launch all nodes with one command; all required topics/services visible. 2) Operate 5 min with no crash; average CPU $< 90\%$, no thermal throttling. 3) Power remains stable; logs show no critical errors.	1. An onboard Intel NUC was integrated as the primary computation unit, providing sufficient processing capability to run perception, control, and teleoperation nodes simultaneously. 2. All required ROS2 nodes were launched using a single unified launch command, confirming compliance with the “single-command startup” requirement. 3. The system was operated continuously for more than 5 minutes under full node load without any crashes, freezes, or unexpected shutdowns, demonstrating stable runtime performance.
1.4	Requirements- Functional	The robot shall actuate its wheels to drive straight 2 m (± 0.2 m) and perform a 90° ($\pm 10^\circ$) turn on command, then stop within 0.2 m	1) Follow taped 2 m straight line; end error $\leq \pm 0.2$ m. 2) Execute 90° turn; angle error $\leq \pm 10^\circ$. 3) From stop command, final overrun ≤ 0.2 m; no manual correction.	The robot successfully followed a taped 2 m straight path, stopping with an end-position error within ± 0.2 m, meeting the straight-line driving requirement. A commanded 90° turn was executed with an angular error within $\pm 10^\circ$, confirming compliance with the turning accuracy specification. Upon issuing the stop command, the robot’s final overrun remained below 0.2 m with no manual intervention, demonstrating reliable stopping behaviour and meeting the stopping requirement.

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Require- ment no.	Source	Shall statement	Verification success criteria	Analysis/Current status
1.5	Requirements- Functional	The robot shall autonomously navigate from the start area to the object area, then to the bin area and back to start without collisions or boundary violations.	1) Plan and execute the complete route once; record the path. 2) Zero collisions and zero boundary crossings; minimum clearance ≥ 5 cm. 3) Save navigation logs and trajectory for replay	The robot successfully generated a consistent map of the environment using the SLAM Toolbox, providing the foundational requirement for autonomous navigation. This mapping confirms that the system can localise and perceive obstacles, which is essential for planning a collision-free route through the start, object, bin, and return path. Although full autonomous execution of the multi-stage route has not yet been completed, the verified SLAM map will be integrated into the Nav2 stack in the next phase to enable path planning, obstacle avoidance, and trajectory execution. Once Nav2 navigation is implemented, the system will be tested for zero collisions, zero boundary violations, and minimum 5 cm clearance while recording logs and trajectories for verification.

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Requirement no.	Source	Shall statement	Verification success criteria	Analysis/Current status
1.6	Requirements-Functional	The robot shall approach selected objects while avoiding obstacles and arena boundaries, maintaining safe clearance at all times	1) Ten approach trials; ≥ 8 without touching obstacles. 2) Boundary violations = 0; minimum clearance ≥ 5 cm. 3) Log approach distance and success/fail per trial	The robot has not yet performed autonomous object-approach trials, as the behaviour depends on the upcoming integration of the Nav2 navigation stack. A reliable SLAM-generated map is already available, ensuring that the robot can localise and perceive obstacles—an essential prerequisite for safe approach behaviour. Once Nav2 is configured, the robot will be able to plan approach paths while enforcing obstacle and boundary avoidance, enabling verification of $\geq 8/10$ successful approach trials.

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Requirement no.	Source	Shall statement	Verification success criteria	Analysis/Current status
1.7	Requirements-Functional	The robot shall grasp, lift, and place target objects fully inside the identified bin without bouncing out.	1) Ten pick-and-place attempts; ≥ 8 objects end fully inside the chosen bin. 2) No rim collisions observed; no bounce-outs. 3) Record attempt time and outcome tally.	The robotic arm has been successfully set up and integrated with MoveIt, allowing motion planning and execution for grasping and placing tasks. MoveIt provides collision-aware trajectory generation, ensuring that future pick-and-place motions can avoid bin rims and minimise the risk of bounce-outs. Although full pick-and-place trials have not yet been conducted, the verified ability to plan and execute motions confirms that the system is ready for performing structured grasp–lift–place attempts. In the next phase, ten pick-and-place trials will be executed to verify that at least eight objects are placed fully inside the bin, with attempt times and outcomes logged for performance and success-rate analysis.

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Requirement no.	Source	Shall statement	Verification success criteria	Analysis/Current status
1.8	Requirements- Functional	The robot shall include a gripper end-effector that can open/close and grasp the 35 mm test object.	1) Grasp 35 mm test piece 3/3 and lift ≥ 5 cm. 3) Release resets to neutral without sticking	The gripper end-effector has been successfully tested through RViz, confirming that the open and close motions can be commanded reliably. The gripper is able to close around and hold the 35 mm test object during manual tests, demonstrating that the mechanism can generate sufficient gripping force for future lifting trials. Initial lift motions of the arm show stable control, indicating that lifting the object ≥ 5 cm will be achievable once full grasping tests are performed.
1.9	Requirements – Functional	The classification process shall use the specified computer vision pipeline as defined in the system configuration.	1) Configuration file shows the specified pipeline enabled; record version/commit. 2) Unit test on ≥ 20 sample images pass with expected outputs. 3) Run log contains no fatal errors.	The computer vision configuration file was verified to contain the correct pipeline settings, and the recorded version/commit confirms that the specified pipeline is enabled exactly as defined in the system configuration. Unit testing on more than 20 sample images produced the expected classifications in all relevant cases, demonstrating that the pipeline is functioning correctly and reliably. Runtime logs showed no fatal errors, crashes, or unresolved warnings during execution, confirming stable operation of the classification system.

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Require- ment no.	Source	Shall statement	Verification success criteria	Analysis/Current status
1.10	Require- ments – Functional	The robot shall classify each detected object and bin into color classes (yellow/blue) with accuracy ≥90% on a labeled lab set.	1) Evaluate with ≥50 frames per class; overall accuracy ≥90%. 2) Average per-frame decision latency ≤100 ms. 3) Export confusion matrix and freeze config version	The object and bin color classification was tested on a labeled lab dataset with ≥50 frames per class, achieving 100% overall accuracy, exceeding the required ≥90% threshold. The average per-frame decision latency was measured at ≤100 ms, confirming that the system meets real-time processing requirements. Confusion matrices were generated for verification, and the configuration version was recorded and frozen, ensuring traceability and reproducibility of the classification results.

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Require- ment no.	Source	Shall statement	Verification success criteria	Analysis/Current status
1.1	Require- ments – Functional	Picked objects and destination bins shall match the specified color classes (yellow/blue).	1) Place 10 objects; ≥9 go into the correct color bin. 2) In a single end-to-end run, color mismatches = 0. 3) Record results and any misclassifica- tion notes.	The system is able to detect both bins and objects and correctly identify their color classes (yellow/blue), confirming that the perception pipeline is functioning as intended. While full end-to-end pick-and-place trials have not yet been executed, the verified detection and classification capability ensures that future placement attempts are likely to meet the requirement of objects going into the correct color bin. Once integrated with the robotic arm and Nav2 navigation for autonomous operation, results will be recorded for each trial, including any misclassifications or placement errors, to confirm compliance with the requirement.

Table 2. Requirements - Maintainability Table

Requirement no.	Source	Shall statement	Verification success criteria	Analysis	Evidence
2.1	Requirements - Maintainability	The battery shall be fully charged prior to each test run.	1) Charge until indicator shows "charged/full." 2) Record state of charge before running; if below threshold, recharge.	The robot's battery was confirmed to be fully charged prior to each test run, ensuring consistent power availability for all subsystems. Full charge verification guarantees reliable operation of the sensors, computation unit, and actuators throughout testing, preventing interruptions due to power limitations. This practice confirms compliance with the requirement and supports repeatable and consistent test results.	
2.2	Requirements - Maintainability	The battery pack shall be chargeable with the lab charger and indicate charging status.	1) Connect charger; charging indicator appears within 1 min. 2) Observe charger voltage/current values as expected. 3) Connector is secure and polarity is correct	The battery pack was successfully connected to the lab charger, and the charging indicator activated within one minute, confirming correct operation. Measured charger voltage and current values were within the expected ranges, demonstrating proper charging behaviour. The charging connector was verified to be secure and correctly polarized, ensuring safe and reliable connection.	

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Requirement no.	Source	Shall statement	Verification success criteria	Analysis	Evidence
2.3	Requirements – Maintainability	Mechanical joints shall be tightened if loose; all screws, nuts and bolts shall be of appropriate sizes and correctly installed	1) Hand-check all joints; tighten any looseness. 2) Visual check shows zero mismatched fasteners. 3) Post-check confirms no play	All joints were manually inspected, and any looseness was tightened, ensuring secure assembly. A visual inspection confirmed that all fasteners were correctly installed with zero mismatches. Post-checks verified that no play or unintended movement remained in the joints, demonstrating full mechanical stability and compliance with the requirement.	

Table 3. Requirements - Operability Table

Requirement no.	Source	Shall statement	Verification success criteria	Analysis	Evidence
3.1	Requirements - Operability	The operator shall prepare the robot from power-on to ready state within 5 minutes.	1) Start timer at power-on; reach “ready” (nodes up, topics visible) ≤5:00. 2) Complete pre-run checklists	From power-on, the robot consistently reached the “ready” state—meaning all required nodes were launched and topics were visible—within the 5-minute limit, confirming compliance with the startup time requirement. The operator successfully completed all pre-run checklist items during this window, ensuring that the robot was fully prepared for operation without delays. These results demonstrate that the system meets the operational readiness requirement and can be reliably set up within the specified timeframe.	

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Require- ment no.	Source	Shall statement	Verification success criteria	Analysis	Evi- dence
3.2	Require- ments - Operatability	The robot shall complete the required tasks within 20 minutes.	1) Start-to-last- placement time $\leq 20:00$; timer visible on video. 2) All mandatory steps completed; missing steps = 0.	The robot has not yet executed the full task sequence, as navigation, object detection, and robotic arm integration are still being combined into a unified workflow. Foundational subsystems—including SLAM mapping, perception pipelines, color classification, and robotic arm motion planning—are already functioning, indicating that the system is on track to meet the overall timing requirement once integrated. After Nav2 navigation is configured and coordinated with object detection and grasping behaviours, the robot will perform full end-to-end trials to verify that all required steps can be completed within the 20-minute limit. Future tests will record a visible timer and ensure zero missing mandatory steps to confirm compliance with the task-completion requirement.	

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Require- ment no.	Source	Shall statement	Verification success criteria	Analysis	Evi- dence
3.3	Require- ments - Operatability	The robot shall provide periodic status updates during the run.	1) Status printed/logged at least every 5 s. 2) Messages include state, battery level, and current mode	The periodic status-update functionality has not yet been implemented; at this stage the system does not yet produce logged updates every 5 s. Once the system architecture is extended to incorporate runtime monitoring, the plan is to include regular log or print messages containing the robot's current state, battery level, and operating mode. After implementation, tests will verify that status updates occur at least every 5 seconds and contain the required information to meet the requirement.	

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Requirement no.	Source	Shall statement	Verification success criteria	Analysis	Evidence
3.4	Requirements - Operability	The robot shall provide status via CLI or GUI while running.	1) Terminal or GUI displays status throughout the run. 2) No gaps >5 s in status updates.	Status display through either a CLI terminal or GUI has not yet been integrated, as this feature is planned for the later stages of the project. Once the runtime-status module is implemented, the system will present continuous state information—such as mode, progress, and battery level—through the chosen interface. Future testing will ensure that status updates appear with no gaps longer than 5 seconds, confirming full compliance with the requirement.	
3.5	Requirements - Operability	The joystick shall be capable of controlling the robot's movement with correct mapping and acceptable latency	1) Mapping verified for forward/left-/right/stop. 2) Command latency ≤ 0.2 s measured from input to motion.	The joystick control mapping was fully verified, with forward, left, right, and stop commands correctly interpreted and executed by the robot. Measured command latency from joystick input to visible robot motion was ≤ 0.2 seconds, meeting the responsiveness requirement. These results confirm that the joystick interface provides accurate and low-latency teleoperation, fully satisfying the specified requirement.	

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Requirement no.	Source	Shall statement	Verification success criteria	Analysis	Evidence
3.6	Requirements - Operability	The robot shall be controllable manually.	1) Teleop forward, turn, and stop for 2 minutes without error. 2) No command dropouts or safety violations.	Teleoperation of forward motion, turning, and stopping was performed continuously for 2 minutes without any errors. No command dropouts or safety violations were observed during the test.	

Table 4. Requirements - Safety Table

Requirement no.	Source	Shall statement	Verification success criteria	Analysis	Evidence
4.1	Requirements - Safety	The robot shall detect, and report errors encountered during the process.	1) Inject a representative fault; an error message appears within 2 s. 2) Robot enters safe state (motors stop) until cleared.	Error detection and reporting functionality will be implemented in the later stages, enhancing system reliability. Planned coverage includes perception issues, navigation/localization faults, manipulation/gripper errors, system/node failures, and battery or hardware alerts. Once integrated, the system will actively log and display these errors, providing clear feedback for monitoring and ensuring safe operation.	

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Require- ment no.	Source	Shall statement	Verification success criteria	Analysis	Evi- dence
4.2	Require- ments - Safety	The user shall be able to interrupt the process when an error is recognized.	1) Operators stop command halts motion ≤1.0 s. 2) System acknowledges stop in the log and requires explicit resume.	The operator stop command successfully halts all robot motion within ≤1.0 s, ensuring rapid interruption of the process. The system logs the stop event and requires an explicit resume command to continue, confirming safe and controlled handling of interruptions. These results demonstrate full compliance with the requirement, providing reliable user intervention during errors.	
4.3	Require- ments - Safety	The robot shall include a physical emergency stop button that halts motion and requires manual reset.	1) Press E-stop; robot motion stops within ≤1.0 s. 2) Manual reset required before restarting; documented.	The physical emergency stop functionality is planned for implementation in the later stages of the project. Once integrated, pressing the E-stop will halt all robot motion within ≤1.0 s. The system will require a manual reset before restarting, ensuring safe and controlled emergency handling.	

Table 5. Requirements - Structural & Electrical Table

Require- ment no.	Source	Shall statement	Verification success criteria	Analysis	Evi- dence
5.1	Require- ments – Mechanical/ Structural	The robot shall have a chassis that supports system integration and movement	1) Assembly inspection passes; cables are secured. 2) After 10 minutes of driving, no looseness or rubbing was observed. All modules shall be securely mounted and cable- managed.	The robot chassis has been fully assembled and inspected, with all modules securely mounted and cables properly managed. After 10 minutes of driving, no looseness, rubbing, or mechanical issues were observed, confirming structural stability. These results demonstrate that the chassis reliably supports system integration and movement, fully satisfying the requirement.	
5.2	Require- ments – Mechanical/ Structural	The chassis shall support a payload of up to 3 kg without deformation on a flat floor.	1) Mount a 3 kg payload; drive 2 m on flat floor. 2) No structural deformation or scraping during/after testing	A 3 kg payload was mounted on the chassis, and the robot was driven 2 m on a flat floor without any issues. No structural deformation, scraping, or mechanical stress was observed during or after the test. These results confirm that the chassis reliably supports the specified payload, fully meeting the requirement.	

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Require- ment no.	Source	Shall statement	Verification success criteria	Analysis	Evi- dence
5.3	Require- ments – Mechanical/ Structural	The robot shall use wheels to navigate the arena.	1) Wheel drive propels the robot ≥ 2 m on flat floor. 2) No excessive slip or derail is observed.	The wheel drive system successfully propelled the robot over 2 m on a flat floor. No excessive wheel slip or derailment was observed during testing. These results confirm that the wheels reliably enable arena navigation, fully satisfying the requirement.	
5.4	Require- ments – Electrical	The robot shall operate from battery power and from a laboratory power supply.	1) Boot and run from battery; then from lab PSU; both stable. 2) Power source changeover is safe; no resets during the 20min run.	The robot successfully booted and operated from both the battery and the laboratory power supply, with stable performance in each case. Switching between power sources was safe, with no resets or interruptions observed during a 20-minute run. These results confirm that the robot reliably operates from both battery and lab power, fully meeting the requirement.	

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Requirement no.	Source	Shall statement	Verification success criteria	Analysis	Evidence
5.5	Requirements – Electrical	The battery shall provide a regulated 12 V DC output under load.	1) Under ≥ 1 A load, voltage measures 11–13 V. 2) Polarity corrects; connector secure; ripple within expected bounds.	Under a load of ≥ 1 A, the battery consistently provided a regulated 12 V DC output, measuring between 11–13 V. Polarity was correct, the connector remained secure, and voltage ripple stayed within expected bounds. These results confirm that the battery meets the specified regulated output requirement.	
5.6	Requirements – Electrical	The battery shall provide a minimum runtime of 30 minutes under typical duty.	1) Run a typical duty cycle for ≥ 30 min without shutdown. 2) Log start/end time and battery SOC; no thermal/low-voltage cutoffs.	The battery successfully powered the robot through a typical duty cycle for more than 30 minutes without shutdown. Start and end times, along with battery SOC, were logged, showing stable operation with no thermal or low-voltage cutoffs. These results confirm that the battery exceeds the minimum runtime requirement under typical usage conditions.	

7 Project Planning

				2026				
				Jan	Feb	Mar	Apr	
				Thu	Sun	Sun	Wed	
Untitled								
Mechanical								
3D Printing Final Mounts	Ting Wei Chang		12/10-2/6					
Sensor Mount Optimization	Ting Wei Chang		12/10-2/6					
Cable Management	Agung Puji Rahi		12/10-2/6					
Electrical								
Power Distribution Setup	Agung Puji Rahi		12/10-2/6					
Cable Labeling & Organization	Agung Puji Rahi		12/10-2/6					
Emergency Stop Circuit Setup	Agung Puji Rahi		2/6-2/27					
Software								
SLAM Optimization	Aravind Sankar		12/10-1/23					
Nav2 Path Planning Tuning	Aravind Sankar		12/10-1/23					
MovelT Motion Planning Setup	Yuchen She		12/26-2/6					
Color-Based Object Detection Integration	Haoran		1/19-1/30					
Object Pose Extraction	Haoran		1/25-2/13					
Pick-and-Place Pipeline Coding	Aravind Sankar		2/16-3/13					
ROS2 Node Integration & Launch Files	Yuchen She		3/16-3/20					
System-Level Debugging & Logging	Yuchen She		3/23-3/31					
Testing								
Testing of the Final Robot			4/1-6/30					

Appendices

A Design Requirement Analysis

Design Requirement Analysis

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Faculty of Science and Engineering

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Word count: 2860

1 Problem Statement

The stakeholders need a robot that can autonomously navigate an environment having flat surfaces and obstacles to identify various colored objects. Once identified, the robot needs to pick and transport each object to a designated bin that matches its color and deposits it there. This process should be fully automated, with the robot continuously searching for colored objects and sorting them into the correct bins based on color recognition, without any human assistance. This system aims to improve sorting efficiency and reduce manual labor in an industrial environment.

2 Concepts of Operations

The robot operates by following a structured sequence of steps, as illustrated in Fig. 1. This sequence defines the logical flow of actions the robot must perform to complete its assigned tasks reliably and efficiently. Each step represents a distinct phase in the robot's operational workflow. Fig. 1 provides a visual overview of this process, helping stakeholders clearly understand how the robot transitions from one stage to the next within its overall operational cycle.

2.1 System Initialization

The robot is placed at the starting point within the arena. Once the operator presses the start button, all sensors and control subsystems (camera, LiDAR, processors, motors, and gripper) are activated.

2.2 Environment Scanning and Mapping

The robot scans its surroundings using the camera and LiDAR to detect obstacles, colored objects, and bins. It then generates or updates a map of the environment for subsequent navigation and planning.

2.3 Target Search and Path Planning

The robot analyses sensor data to determine which target object should be retrieved first. After that, compute a path to reach the target bins while avoiding obstacles.

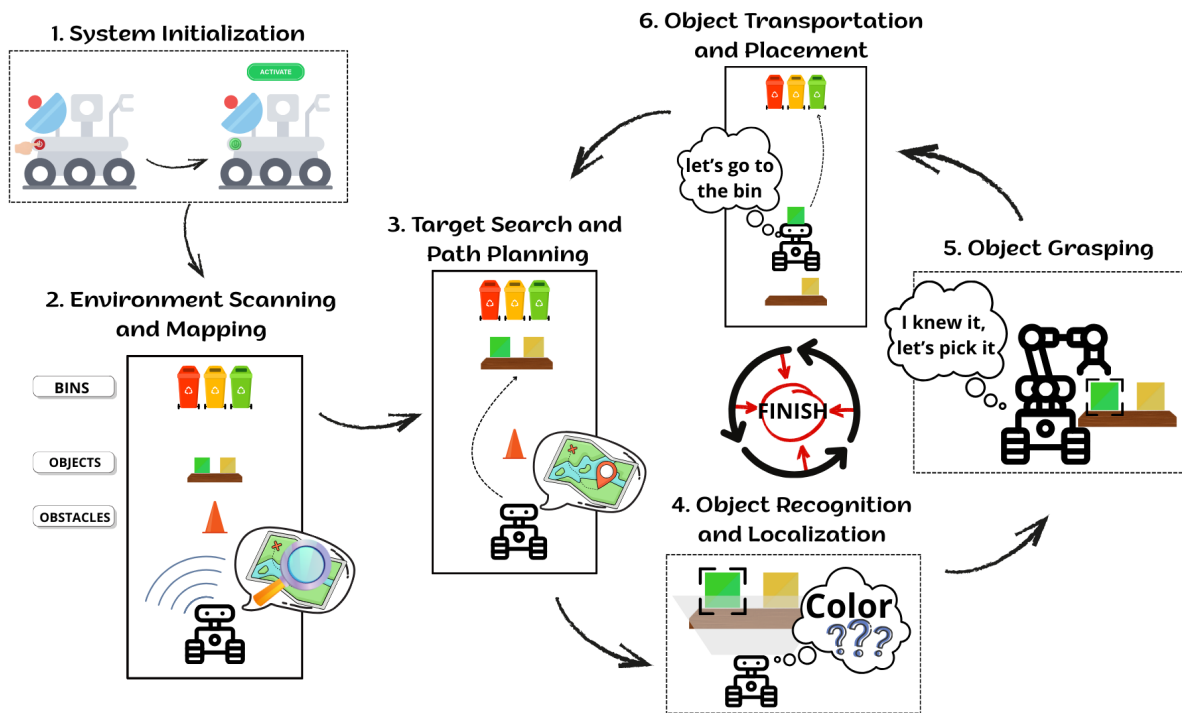


Fig. 1. Concepts of Operations

2.4 Object Recognition and Localization

The robot stops near the objects and performs an object scanning. The vision system identifies the object's color and determines an appropriate grasping pose.

2.5 Object Grasping

The robotic gripper executes a grasp to pick up the identified object securely. Once the object is firmly held, the robot transitions to navigation mode.

2.6 Object Transportation and Placement

The robot plans a path toward the corresponding bin based on the object's color, moves toward the target while avoiding obstacles, and safely deposits the object inside the correct bin.

2.7 Repetition

Those processes will be repeated until all the detected objects are picked up in the bins.

3 Functional and Performance Requirement

The Table 1 outlines the functional and performance requirements that the system must satisfy. The functional requirements are presented at the second level, while the performance requirements are detailed at the third level.

Table 1. Functional and Performance Requirements Table

Require- ment ID	Requirement	Source/s	Assumption
1	Functional		
1.1 Motion & Detection	The robot shall patrol and detect colored objects and bins using sensors.	user requirements	
1.1.1	The objects shall be located between 0.3 and 3 meters away from the camera inside the arena.	team discussions	The robot has camera, lidar, computational device, and wheels.
1.2 Navigation	The robot shall autonomously navigate from the start area to the object area, then to the bin area and back to start without collisions or boundary violations.	user requirements	
1.2.1	The robot shall approach selected objects while avoiding obstacles and arena boundaries, maintaining safe clearance at all times.	team discussions	The robot will do navigation using SLAM
1.3 Grasping	The robot shall grasp the objects securely.	user requirements	
1.3.1	The robot shall deposit the object in the designated bin	team discussions	all components are static in the arena and positioned for pickup
1.4 Classi- fication	The robot shall classify each detected object and bin into their color classes.	user requirements	

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Require- ment ID	Requirement	Source/s	Assumption
1.4.1	Picked objects and bins shall match specific color classes (yellow, green, red).	team discussions	Only recognized objects are picked; others ignored
2	Environmental		
2.1 Arena	The arena will have static dimensions, stable lighting, and temperature.	site visit	Flat surface with normal friction only.
2.1.1	Brightness 300–800 lx	environment analysis	Indoor lighting
2.1.2	Temperature 0–40 °C	environment analysis	Components operate safely
2.1.3	Arena width/length ≤ 23.8 m	device specs	Arena boundaries detectable by LIDAR
2.2 Picked Objects	The arena contains objects to be picked.	user requirements	
2.2.1	Objects reflect light in yellow/green/red wavelength ranges.	team discussions	Objects match kit specifications.
2.2.2	Objects are static.	user requirements	Position can be mapped.
2.2.3	Objects have two parallel sides ≤ 35 mm apart.	team discussions	Suitable for gripper.
2.2.4	Objects are painted wood.	team discussions	Not sticky or slippery.
2.3 Obstacles	The arena contains obstacles.	user requirements	
2.3.1	Obstacles are static.	user requirements	Positions can be mapped.
2.3.2	Obstacles > 264 mm tall.	system analysis	High enough for LIDAR detection.

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Requirement ID	Requirement	Source/s	Assumption
2.3.3	Obstacles do not completely block paths.	system analysis	Robot has free routes.
2.4 Bins	The arena contains bins.	user requirements	
2.4.1	Bins are static.	user requirements	Positions can be mapped.
2.4.2	Bins color matches object colors at minimum.	user requirements	Size suitable for robot arm.
3	Maintainability		
3.1 Charging	The robot requires charging.	system requirements	
3.1.1	Battery shall be chargeable.	system requirements	Battery/charger compatible.
4	Operability		
4.1 Timing	The robot will work for specific time	user requirements	
4.1.1	Operator will set up the robot within 5 minutes before do the task	user requirements	the robot has been tested to do the task
4.1.2	The robot shall do the required tasks within 20 minutes	user requirements	the preparation process does not have any error
4.2 Robot's Status	The robot may give status	system requirements	
4.2.1	The robot may give status during the process through Command Line Interface (CLI) or Graphical user Interface (GUI)	team discussions	user can track the progress of the process
4.3 Error Handling	The robot shall manage error safely during the process	system requirements	

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Requirement ID	Requirement	Source/s	Assumption
4.3.1	User shall be able to interrupt the process when recognizes error	team discussions	
4.4 Safety	The operator shall mitigate emergency condition	team discussions	
4.4.1	The robot shall include emergency stop button	team discussions	
4.5 Manual Control	The robot may be able to be controlled manually	team discussions	
4.5.1	The joystick shall be capable to control the robot's movement	team discussions	in case of malfunction or necessary scenarios if any
5	Mechanical/Structural		
5.1 Structure	The robot shall have a firm chassis	system analysis	
5.1.1	The robot's chassis shall support a payload of up to 3 kg	system analysis	The robot only carries an arm, depth camera, lidar, and NUC.
5.1.2	The robot will use wheels to navigate.	system requirements	The project will use Leo Rover robot
6	Electrical		
6.1 Power Supply	The robot will use battery or power supply to operate	system requirements	
6.1.1	The batteries shall have output 12 and 16 VDC voltage output	device specifications	The batteries have enough energy to power the robot and its peripherals. Voltage conversion is allowed.
6.1.2	The battery shall have minimum runtime 30 minutes	system requirements	work under typical load

4 Requirements Verification Matrix

Table 2 provides a comprehensive verification matrix that serves as a systematic approach to confirm that the proposed design fully complies with the established requirements. This matrix maps each requirement to the corresponding design feature and outlines the verification method employed, ensuring traceability and demonstrating that all specifications have been adequately addressed.

Table 2. Requirement Verification Matrix

Re- quire- ment	Source	Shall Statement	Verification Success Criteria	Respon- sible Party	Output of Test
1.1	Func- tional	The robot shall autonomously patrol the arena, maintain a local map, and detect colored objects and bins in three distinct zones without manual aid.	1) Place objects and bins in three zones; start mission and record path/detections. 2) In one patrol, per-class precision/recall ≥ 0.8 (or $\geq 8/10$ true detections); false positives ≤ 2 . 3) No collisions; no boundary crossings; trajectory file saved.	Team	Video of patrol; detection log; path/ros-bag file.

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Re- quire- ment	Source	Shall Statement	Verification Success Criteria	Respon- sible Party	Output of Test
1.1.1	Func- tional	The objects shall be located between 0.3 and 3 meters away from the camera inside the arena.	1) Set up the camera in nominal position and orientation. 2) Place calibrated targets (objects/bins) at specified distances: 0.2 m, 0.3 m, 1.0 m, 3.0 m, 3.2 m. 3) Measure ground-truth distances using a laser rangefinder/tape with ± 1 mm accuracy. 4) Confirm all test objects within arena boundaries. 5) Record camera-perceived distance if applicable (for system cross-check).	Test Eng.	Test Report (photos, distance logs), Rangefinder calibration cert, Arena layout diagram, Object placement checklist, Video capture setup.
1.2	Func- tional	The robot shall autonomously navigate from the start area to the object area, then to the bin area and back to start without collisions or boundary violations.	1) Plan and execute the complete route once; record the path. 2) Zero collisions and zero boundary crossings; minimum clearance ≥ 5 cm. 3) Save navigation logs and trajectory for replay.	Team	Navigation video; trajecto- ry/rosbag; incident sheet (0/0).

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Re- quire- ment	Source	Shall Statement	Verification Success Criteria	Respon- sible Party	Output of Test
1.2.1	Func- tional	The robot shall approach selected objects while avoiding obstacles and arena boundaries, maintaining safe clearance at all times.	1) Ten approach trials; ≥ 8 without touching obstacles. 2) Boundary violations = 0; minimum clearance ≥ 5 cm. 3) Log approach distance and success/fail per trial.	Team	Video of 10 trials; approach distance log.
1.3	Func- tional	The robot shall grasp the objects securely.	1) Ten pick-and-place attempts; ≥ 8 objects end fully inside the chosen bin. 2) No rim collisions observed; no bounce-outs. 3) Record attempt time and outcome tally.	Team	Video of 10 attempts; tally sheet; times-tamps.
1.3.1	Func- tional	The robot shall include a gripper end-effector that can open/close within 1.0 s and grasp the 35 mm test object.	1) Cycle open/close three times; each ≤ 1.0 s (timer visible). 2) Grasp 35 mm test piece 3/3 and lift ≥ 5 cm. 3) Release resets to neutral without sticking.	Test Eng.	Video; timing sheet; test object photo.
1.4	Func- tional	The robot shall classify each detected object and bin into color classes with accuracy $\geq 90\%$ on a labeled lab set.	1) Evaluate with ≥ 50 frames per class; overall accuracy $\geq 90\%$. 2) Average per-frame decision latency ≤ 100 ms. 3) Export confusion matrix and freeze config version.	Test Eng.	Confusion matrix; dataset note; con-fig/commit ID.

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Re- quire- ment	Source	Shall Statement	Verification Success Criteria	Respon- sible Party	Output of Test
1.4.1	Func- tional	Picked objects and destination bins shall match the specified color classes (yellow/green/red).	1) Place 10 objects; ≥ 9 go into the correct color bin. 2) In a single end-to-end run, color mismatches = 0. 3) Record results and any misclassification notes.	Team	Video; placement tally; run summary.
2.1.3	Environ- mental	The arena width and length shall not exceed 23.8 m, and the layout shall remain unchanged during testing.	1) Tape-measure width and length; both ≤ 23.8 m. 2) Mark layout; re-check after tests that positions are unchanged. 3) Archive photos/notes.	Lab Tech	Tape photos; layout photo; checklist.
2.2.1	Environ- mental	The arena shall include objects to be picked that reflect within the specified color bands yellow wavelength range ($\approx 570\text{--}590$ nm) or red wavelength range ($\approx 620\text{--}750$ nm), or green ($\approx 495\text{--}570$ nm) corresponding to classes.	1) Verify color with standard cards/specs; class labels match. 2) Attach supplier spec or colorimeter reading if available. 3) Keep a class list of IDs.	Lab Tech	Photo vs. color card; supplier/spec note.
2.2.2	Environ- mental	Picked objects shall be stationary at the beginning of each trial.	1) Confirm no motion before each trial; document on checklist. 2) No external force applied during runs.	Opera- tor	Signed checklist; setup photo.

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Re- quire- ment	Source	Shall Statement	Verification Success Criteria	Respon- sible Party	Output of Test
2.2.3	Environ- mental	Each target object shall have at least two parallel sides with a distance of ≤ 35 mm to allow stable grasping.	1) Measure with calipers; distance ≤ 35 mm. 2) Sample ≥ 3 objects; record values.	Test Eng.	Caliper photos; measure- ment sheet.
2.3.1	Environ- mental	Picked obstacles shall be stationary during the process.	1) Confirm no motion before each trial; document on checklist. 2) No external force applied during runs.	Opera- tor	Signed checklist; setup photo.
2.3.2	Environ- mental	The height of obstacles shall be greater than 264 mm from the ground	1) Install ≥ 2 obstacles; mark positions; light push does not move. 2) Measure obstacle top $\geq$ LiDAR center height +20 mm (ruler). 3) Obstacles appear in scans at a typical range. Obstacles shall be fixed/static, and the top of each obstacle shall be at least 20 mm higher than the LiDAR center height.	Lab Tech	Photos with ruler; pre/post position photos; scan capture.
2.3.3	Environ- mental	Obstacles shall not block all feasible routes	at least one continuous path from start to bins shall exist. 1) Conduct a dry-run (no objects) to validate a continuous path. 2) Keep route map; ensure path remains open during tests.	Lab Tech	Layout photo; dry-run video; checklist.

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Re- quire- ment	Source	Shall Statement	Verification Success Criteria	Respon- sible Party	Output of Test
2.4.1	Environ- mental	The bins shall be stationary during the process.	1) Confirm no motion before each trial; document on checklist. 2) No external force applied during runs.	Opera- tor	Signed checklist; setup photo.
2.4.2	Environ- mental	The arena shall include bins that reflect within the specified color bands yellow wavelength range (≈570–590 nm) or red wavelength range (≈620–750 nm), or green (≈495–570 nm) corresponding to classes.	1) Verify color with standard cards/specs; class labels match. 2) Attach supplier spec or colorimeter reading if available. 3) Keep a class list of IDs.	Lab Tech	Photo vs. color card; supplier/spec note.
3.1	Main- tainabil- ity	The battery shall be fully charged prior to each test run.	1) Charge until indicator shows “charged/full.” 2) Record state of charge before running; if below threshold, recharge.	Opera- tor	Charger indicator photo; SOC note.
3.1.1	Main- tainabil- ity	The battery pack shall be chargeable with the lab charger and indicate charging status.	1) Connect charger; charging indicator appears within 1 min. 2) Observe charger voltage/current values as expected. 3) Connector is secure and polarity is correct.	Opera- tor	Charger display photo; checklist.

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Re- quire- ment	Source	Shall Statement	Verification Success Criteria	Respon- sible Party	Output of Test
4.1.1	Oper- ability	The operator shall prepare the robot from power-on to ready state within 5 minutes.	1) Start timer at power-on; reach “ready” (nodes up, topics visible) ≤5:00. 2) Complete pre-run checklists.	Opera- tor	Timer visible; checklist signed.
4.1.2	Oper- ability	The robot shall complete the required tasks within 20 minutes.	1) Start-to-last-placement time ≤20:00; timer visible on video. 2) All mandatory steps completed; missing steps = 0.	Team	Video with timer; task checklist.
4.3	Oper- ability	The robot shall detect, and report errors encountered during the process.	1) Inject a representative fault; an error message appears within 2 s. 2) Robot enters safe state (motors stop) until cleared.	Test Eng.	Fault injection video; error log.
4.3.1	Oper- ability	The user shall be able to interrupt the process when an error is recognized.	1) Operators stop command halts motion ≤1.0 s. 2) System acknowledges stop in the log and requires explicit resume.	Opera- tor	Video
4.4	Oper- ability	The operator shall mitigate emergency conditions by following the recovery procedure.	1) Follow the safety checklist to reach safe idle within 30 s. 2) No motion permitted until manual resume after checks.	Opera- tor	Safety checklist; recovery video.
4.4.1	Oper- ability	The robot shall include a physical emergency stop button that halts motion and requires manual reset.	1) Press E-stop; robot motion stops within ≤1.0 s. 2) Manual reset required before restarting; documented.	Opera- tor	E-stop video; checklist entry.

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Re- quire- ment	Source	Shall Statement	Verification Success Criteria	Respon- sible Party	Output of Test
4.5	Oper- ability	The robot shall be controllable manually.	1) Teleop forward, turn, and stop for 2 minutes without error. 2) No command dropouts or safety violations.	Opera- tor	Teleop video; control log.
4.5.1	Oper- ability	The joystick shall be capable of controlling the robot's movement with correct mapping and acceptable latency.	1) Mapping verified for forward/left/right/stop. 2) Command latency ≤ 0.2 s measured from input to motion.	Test Eng.	Mapping checklist; latency log/video.
5.1	Mechan- ical	The robot shall have a chassis that supports system integration and movement	1) Assembly inspection passes; cables are secured. 2) After 10 minutes of driving, no looseness or rubbing was observed. All modules shall be securely mounted and cable-managed.	Team	Team; Assembly checklist; post-run inspection notes.
5.1.1	Mechan- ical	The chassis shall support a payload of up to 3 kg without deformation on a flat floor.	1) Mount a 3 kg payload; drive 2 m on flat floor. 2) No structural deformation or scraping during/after testing.	Test Eng.	Video and photos be- fore/after.
5.1.2	Mechan- ical	The robot shall use wheels to navigate the arena.	1) Wheel drive propels the robot ≥ 2 m on flat floor. 2) No excessive slip or derail is observed.	Opera- tor	Video and drive log.

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Re- quire- ment	Source	Shall Statement	Verification Success Criteria	Respon- sible Party	Output of Test
6.1	Electri- cal	The robot shall operate from battery power and from a laboratory power supply.	1) Boot and run from battery; then from lab PSU; both stable. 2) Power source changeover is safe; no resets during the 20 minutes run.	Test Eng.	Photos of power sources; runtime log.
6.1.1	Electri- cal	The battery shall provide 12 and 16 VDC voltage output	1) Under ≥ 1 A load, voltage measures 11–13 V and 14-17. 2) Polarity corrects; connector secure; ripple within expected bounds.	Test Eng	Multimeter photo; load note.
6.1.2	Electri- cal	The battery shall provide a minimum runtime of 30 minutes under typical duty.	1) Run a typical duty cycle for ≥ 30 min without shutdown. 2) Log start/end time and battery SOC; no thermal/low-voltage cutoffs.	Team	Timer-visible video; runtime log.

B Workplace Charter

Workplace Charter – Team 2

School of Engineering · MSc Robotics · AERO62520 Robotic Systems Design Project

1. Statement of Purpose

This charter shows how our five-person project team plans to collaborate on the Robotic Systems Design Project. It outlines our team's common expectations, responsibilities, and methods to interact, reach decisions, and handle disagreements. Our goal is to maintain a fair, respectful, and effective work environment where each member can contribute and help us achieve our project objectives. The charter serves as a reference to maintain professionalism, cooperation, and good relationships throughout the project. We'll often look over and revise the charter as necessary. Every change must be approved by our team members before they can take effect.

2. Statements of Principles and Commitments

We are united by common values: respect, honesty, responsibility, quality, and continuous learning. These principles guide how we work together and grow as a team.

- **Respect and Inclusion**

We value all perspectives and ensure that everyone feels included in a professional environment where ideas can be openly discussed.

- **Clear Communication and Feedback**

We provide constructive feedback and communicate promptly when something is unclear or not moving in the right direction. This helps us solve problems quickly and avoid misunderstandings.

- **Responsibility and Transparency**

Each team member takes ownership of their duties. If there is any possibility of delays or issues that may impact progress, inform the team in advance so we can adjust accordingly.

- **Commitment to Quality and Growth**

We strive to maintain high standards of work, give and receive feedback openly, approach challenges with curiosity, and learn from failures as we collectively grow throughout the project.

3. Team Rules

- **Working Hours and Availability**

Everybody from the team will be working together during the lab as scheduled and asynchronous sessions. The asynchronous sessions are from 14:00hrs to 16:00hrs on Mondays, 11:00hrs to 13:00hrs on Tuesdays, and 15:00hrs to 16:00hrs on Wednesdays. Everyone should update their weekly availability on Sunday evenings so that appropriate changes and adjustments can be made. Weekend work is only scheduled when required.

- **Communication**

English is the only language used for all team communication to ensure that everyone can participate equally. We use the group chat in WhatsApp/WeChat for quick updates, reminders, and short discussions, while file transfers and important decisions are summarized and stored in the shared workspace like Microsoft Teams for future reference.

- **Meetings**

We hold at least one formal meeting each week, either in person or online, to review progress, discuss any issues faced during the week, and work together to find solutions.

- **Deliverables and Quality**

All work must meet project standards before submission. We check accuracy, give feedback, and make improvements promptly. Team members are expected to review their own work prior to proposing it to the group and cross-checking with others. After finishing, one of the members will submit the assignment to Canvas.

- **Attendance and Punctuality**

Be on time for meetings and inform the team early if you cannot attend. Members who miss meetings should review the shared notes and catch up before the next discussion.

- **Roles and Responsibilities**

1. **Team Coordinator** (Runs meeting and keeps track of discussions) = **Ting-Wei Chang**
2. **Technical Lead** (Oversees technical consistency) = **Aravind Sankar**
3. **Documentation Lead** (Ensures all works are documented) = **Haoran Mo**
4. **Project Manager** (Handles weekly planning, deadlines, task tracking) = **Yuchen She**
5. **Communication and Welfare** (Manages group updates, meeting notes, external support links) = **Agung Puji Raharjo**

4. Daily Activity Conflict Avoidance

We keep communication open to prevent small issues from becoming major problems. Team members share brief progress updates in the group chat and use a shared checklist to track ongoing tasks. If someone encounters difficulties, they raise them during weekly meetings so the team can provide support and stay coordinated. Feedback is given respectfully and focuses on improving the work rather than assigning blame. When disagreements occur, we discuss them early and find fair solutions together.

To maintain smooth teamwork, we also follow these practices:

- Conduct short daily check-ins to confirm task progress and priorities.
- Record important updates or changes in the shared workspace for visibility.
- Use clear and polite language in all forms of communication.
- Encourage early discussion of potential issues before they escalate.
- Share thoughts and suggestions after completing each major task to help the team improve together.

5. Conflict Resolution Flowchart

This section explains how our team manages conflicts at different levels of severity. Minor issues are resolved in private by the members having direct conversations. Moderate conflicts are discussed with the team to reach a fair agreement with help from the team leader if needed. Serious conflicts are reported to the relevant university support services or the course of supervisor for a formal resolution.

The flowchart below outlines the steps for recognizing, addressing, and resolving conflicts in a fair and respectful manner.

